Code Coverage Report

Line and function coverage of the server code, measured while running the full automated test suite.

1. Summary

Line coverage	Branch coverage	Function coverage	Tests run
94.37 %	83.91 %	96.72 %	57 (all passing)

Scope: server source in src/ (excluding the generated Prisma client). index.ts (process entry point) and seed.ts (data seeding) are not loaded by the tests and are not in the figures. The React client (client/src) has no automated tests and is not covered. Measured with Node's built-in V8 coverage (--experimental-test-coverage).

2. Coverage by file

File	Line %	Function %	Uncovered lines
activityLog.ts	92.16	100.00	1-2, 14-15
app.ts	73.68	50.00	23, 27, 32-34, 43-52
attempts.ts	93.16	100.00	1-3, 149-151, 182-191, 217-218
auth.ts	94.83	85.71	1-2, 68-71
backup.ts	95.67	100.00	1-6, 51-52, 111-112, 215-216
exams.ts	97.39	100.00	1-2, 140-141
grading.ts	97.85	100.00	1-2
http.ts	86.11	100.00	1, 3, 29-30, 51-52, 55-56, 64-65
prisma.ts	92.31	100.00	9-10
questions.ts	98.90	100.00	1-2
reports.ts	97.70	100.00	1-2
results.ts	98.21	100.00	1-2
routes.ts	98.00	100.00	1
security.ts	83.33	100.00	1-7, 42-44
All files	94.37	96.72	

3. Gaps

The lowest line coverage is in app.ts (73.7%: static-file serving and the catch-all/error handlers at lines 43-52, only reachable with a built client or an unhandled error), security.ts (83.3%) and http.ts (86.1%: unusual error-mapping branches). Most other uncovered lines are import/boilerplate lines or rare error paths. Business logic files (grading, questions, exams, results, reports, routes) are all above 97%.

4. Reproduce

cd online-exam-portal/server && npx tsx --test --test-concurrency=1 --experimental-test-coverage --test-coverage-include="src/**" --test-coverage-exclude="src/generated/**" "test/*.test.ts"